Exercise 1. Prove:

a) $S \cap (H \cup U) = (S \cap H) \cup (S \cap U)$.

b) $S \cup (H \cap U) = (S \cup H) \cap (S \cup U)$.

Proof. Let S, H, U be any sets.

a) We want to prove that $x \in S \cap (H \cup U)$ if and only if $x \in (S \cap H) \cup (S \cap U)$. Then $x \in S \cap (H \cup U) \iff x \in S$ and $x \in H \cup U \iff x \in S$ and $x \in H$ or $x \in U \iff x \in S$ and $x \in H$ or $x \in S$ and $x \in U \iff x \in (S \cap H) \cup (S \cap U)$. Thus, $x \in S \cap (H \cup U)$ if and only if $x \in (S \cap H) \cup (S \cap U)$, as required.

b) We want to prove that $x \in S \cup (H \cap U)$ if and only if $x \in (S \cup H) \cap (S \cup U)$. Then $x \in S \cup (H \cap U) \iff x \in S$ or $x \in H \cap U \iff x \in S$ or $x \in H$ and $x \in U \iff x \in S$ or $x \in H$ and $x \in S$ or $x \in U \iff x \in (S \cup H) \cap (S \cup U)$. Thus, $x \in S \cup (H \cap U)$ if and only if $x \in (S \cup H) \cap (S \cup U)$, as required.

□

Exercise 2. Prove by induction that $n < 2^n$ for all $n \in \mathbb{N}$.

Proof. Prove that $n < 2^n$ for all $n \in \mathbb{N}$ by induction.

1) (Base Case) $1 < 2$.

2) (Inductive Step) Assume $m < 2^m$. Since $m \in \mathbb{N}$, $1 \leq m < 2^m$. Thus,

$$m + 1 < 2^m + 1 < 2^m + 2^m = 2^{m+1},$$

as required.

By induction, $n < 2^n$ for all $n \in \mathbb{N}$, as needed. □

Exercise 3. Show that for a finite set A of cardinality n, the cardinality of the power set $P(A)$ is 2^n.

Proof. Let A be any finite set of size n. We want to prove that the size of the power set $P(A)$ is 2^n by induction.

1) (Base Case) $|A| = 0 \implies A = \emptyset \implies P(A) = \{\emptyset\} \implies |P(A)| = 1$.

2) (Inductive Step) Suppose $|A| = m \implies |P(A)| = 2^m$. If $|A| = m + 1$, then

$$\forall x \in A : |A \setminus \{x\}| = m \implies \forall x \in A : |P(A \setminus \{x\})| = 2^m;$$

hence,

$$\forall x \in A : |P(A)| = |P(A \setminus \{x\} \cup \{x\})| = |P(A \setminus \{x\}) \times P(\{x\})| = 2^m \cdot 2 = 2^{m+1}$$

and $|P(A)| = 2^{m+1}$.

By induction, the size of the power set $P(A)$ is 2^n, as wanted. □

Exercise 4. Prove that $n^3 + 5n$ is divisible by 6 for all $n \in \mathbb{N}$.

Proof. Prove that $2 \mid n^2 + n$ and $6 \mid n^3 + 5n$ for all $n \in \mathbb{N}$ by induction. We need to show that $2 \mid n^2 + n$ for all $n \in \mathbb{N}$.

1) (Base Case) $2 \mid 1 + 1$.

2) (Inductive Step) Assume $2 \mid m^2 + m$. Thus,

$$2 \mid (m^2 + m) + (2m + 2) = m^2 + 3m + 2 = (m + 1)(m + 2) = (m + 1)^2 + (m + 1),$$

as needed.

By induction, $2 \mid n^2 + n$ for all $n \in \mathbb{N}$. We want to show that $6 \mid n^3 + 5n$ for all $n \in \mathbb{N}$.

1) (Base Case) $6 \mid 1 + 5$.

2) (Inductive Step) Assume $6 \mid m^3 + 5m$. Since $2 \mid m^2 + m$, $6 \mid 3m^2 + 3m$. Thus,

$$6 \mid (m^3+5m)+(3m^2+3m)+6 = (m^3+3m^2+3m+1)+(5m+5) = (m+1)^3+5(m+1),$$

as desired.

By induction, $6 \mid n^3 + 5n$ for all $n \in \mathbb{N}$, as wanted. □

Exercise 5. Give an example of a countably infinite collection of finite sets $A_1, A_2, ...$, whose union is an infinite set.

Proof. If $\forall n \in \mathbb{N} : A_n = \{n\} \implies |A_n| = 1$, then

$$\bigcup_{n=1}^{\infty} A_n = \mathbb{N} \implies \left| \bigcup_{n=1}^{\infty} A_n \right| = |\mathbb{N}|,$$

as needed. □

Exercise 6. In this exercise, you will prove that

$$|\{q \in \mathbb{Q} : q > 0\}| = |\mathbb{N}|.$$

Define $f : \{q \in \mathbb{Q} : q > 0\} \to \mathbb{N}$ as follows: $f(1) = 1$, if $q \in \mathbb{N} \setminus \{1\}$ is given by some prime factorization, then

$$q = p_1^{r_1} \cdots p_N^{r_N} \implies f(q) = p_1^{2r_1} \cdots p_N^{2r_N},$$

and if $q \in \mathbb{Q} \setminus \mathbb{N}$ is given by some prime factorization, then

$$q = \frac{m_1^{s_1} \cdots m_N^{s_N}}{n_1^{t_1} \cdots n_M^{t_M}} \implies f(q) = m_1^{2s_1} \cdots m_N^{2s_N} n_1^{2t_1-1} \cdots n_M^{2t_M-1}.$$

a) Compute $f(4/15)$. Find q such that $f(q) = 108$.

b) Prove that f is a bijection.

Proof. By the Fundamental Theorem of Arithmetic,

1) $\forall q \in \mathbb{N} \setminus \{1\} : \exists!$ primes $p_1 < \cdots < p_N \in \mathbb{N}$ AND exponents $r_1, ..., r_N \in \mathbb{N}$ such that

$$q = p_1^{r_1} \cdots p_N^{r_N}. \tag{†}$$

2) $\forall q \in \mathbb{Q} \setminus \mathbb{N} : \exists!$ primes $m_1 < \cdots < m_N \in \mathbb{Z}$ and $n_1 < \cdots < n_M \in \mathbb{N}$ AND exponents $s_1, ..., s_N \in \mathbb{N}$ and $t_1, ..., t_M \in \mathbb{N}$ such that

$$q = \frac{m_1^{s_1} \cdots m_N^{s_N}}{n_1^{t_1} \cdots n_M^{t_M}}. \tag{‡}$$

Define $f : \{q \in \mathbb{Q} : q > 0\} \to \mathbb{N}$ as follows: $f(1) = 1$, $f(q) = p_1^{2r_1} \cdots p_N^{2r_N}$ if $q \in \mathbb{N} \setminus \{1\}$ is given with (†), and $f(q) = m_1^{2s_1} \cdots m_N^{2s_N} n_1^{2t_1 - 1} \cdots n_M^{2t_M - 1}$ if $q \in \mathbb{Q} \setminus \mathbb{N}$ is given with (‡).

a) We want to compute $f(4/15)$. If $q = 4/15$, then $f(q) = 16 \cdot 3 \cdot 5 = 240$. We want to find $q \in \mathbb{Q}$ such that $f(q) = 108$. If $q = 2/9$, then $f(q) = 4 \cdot 27 = 108$.

b) We want to prove that f is an injection. Suppose $f(q_1) = f(q_2)$. If $f(q_1) = 1 = f(q_2)$, then

$$q_1 = 1 = q_2.$$

Assume $f(q_1) \neq 1 \neq f(q_2)$. Using the Fundamental Theorem of Arithmetic, $f(q_1)$ and $f(q_2)$ have identical prime factorizations and hence q_1 and q_2 have similar prime factorizations of either (†) or (‡) in form; otherwise, $f(q_1)$ and $f(q_2)$ have distinct prime factorizations. Without loss of generality, let $m_1 < \cdots < m_N \in \mathbb{N}$ and $n_1 < \cdots < n_M \in \mathbb{N}$ be the unique primes AND $s_1, ..., s_N \in \mathbb{N}$ and $t_1, ..., t_M \in \mathbb{N}$ be the unique exponents such that

$$q_1 = m_1^{s_1} \cdots m_N^{s_N}$$

and

$$q_2 = n_1^{t_1} \cdots n_M^{t_M}.$$

Then

$$q_1^2 = m_1^{2s_1} \cdots m_N^{2s_N} = n_1^{2t_1} \cdots n_M^{2t_M} = q_2^2 \implies q_1 = q_2.$$

Without loss of generality, let $a_1 < \cdots < a_N \in \mathbb{N}$ and $b_1 < \cdots < b_M \in \mathbb{N}$ and $c_1 < \cdots < c_{N'} \in \mathbb{N}$ and $d_1 < \cdots < d_{M'} \in \mathbb{N}$ be the unique primes AND $r_1, ..., r_N \in \mathbb{N}$ and $s_1, ..., s_M \in \mathbb{N}$ and $r'_1, ..., r'_{N'} \in \mathbb{N}$ and $s'_1, ..., s'_{M'}$ such that

$$q_1 = \frac{a_1^{r_1} \cdots a_N^{r_N}}{b_1^{s_1} \cdots b_M^{s_M}}$$

and

$$q_2 = \frac{c_1^{r'_1} \cdots c_{N'}^{r'_{N'}}}{d_1^{s'_1} \cdots d_{M'}^{s'_{M'}}}.$$

Then
$$b_1^{4s_1-1}\cdots b_M^{4s_M-1}q_1^2 = d_1^{4s_1'-1}\cdots d_{M'}^{4s_{M'}'-1}q_2^2 \implies q_1 = q_2.$$
Thus, $q_1 = q_2$ and f is an injection, as desired. We want to prove that f is a surjection. Suppose $p \in \mathbb{N}$. If $p = 1$, then

$$p = f(1) \in f(\{q \in \mathbb{Q} : q > 0\}).$$

Assume $p \neq 1$. Using the Fundamental Theorem of Arithmetic, p can be factored into even or odd powers of unique primes. Without loss of generality, let $p_1 < \cdots < p_N \in \mathbb{N}$ be the unique primes and $r_1, ..., r_N \in \mathbb{N}$ be the unique exponents such that

$$p = p_1^{2r_1}\cdots p_N^{2r_N}.$$

Then
$$p = f(p_1^{r_1}\cdots p_N^{r_N}) \in f(\{q \in \mathbb{Q} : q > 0\}.$$
Without loss of generality, let $m_1 < \cdots < m_N \in \mathbb{N}$ and $n_1 < \cdots < n_M \in \mathbb{N}$ be the unique primes AND $s_1, ..., s_N \in \mathbb{N}$ and $t_1, ..., t_M \in \mathbb{N}$ be the unique exponents such that
$$p = m_1^{2s_1}\cdots m_N^{2s_N}n_1^{2t_1-1}\cdots n_M^{2t_M-1}.$$
Then
$$p = f\left(\frac{m_1^{s_1}\cdots m_N^{s_N}}{n_1^{t_1}\cdots n_M^{t_M}}\right) \in f(\{q \in \mathbb{Q} : q > 0\}.$$
Without loss of generality, let $p_1 < \cdots < p_N \in \mathbb{N}$ be the unique primes and $r_1, ..., r_N \in \mathbb{N}$ be the unique exponents such that

$$p = p_1^{2r_1-1}\cdots p_N^{2r_N-1}.$$

Then
$$p = f\left(\frac{1}{p_1^{r_1}\cdots p_N^{r_N}}\right) \in f(\{q \in \mathbb{Q} : q > 0\}).$$
Thus, $p \in f(\{q \in \mathbb{Q} : q > 0\}$ and f is an injection, as desired.

Thus, $f : \{q \in \mathbb{Q} : q > 0\} \to \mathbb{N}$ is a bijection and

$$|\{q \in \mathbb{Q} : q > 0\}| = |\mathbb{N}|,$$

as wanted. □